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GTIIT

CALCULUS 1 - M

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LECTURE 14 - NOVEMBER 27TH

INTEGRAL: COMPUTATION OF THE AREA UNDER THE CURVE

Goal: To find the area of the region that lies under the curve $y = f(x)$ on $[a, b]$

The problem is easy to answer for **regions with straight sides**:

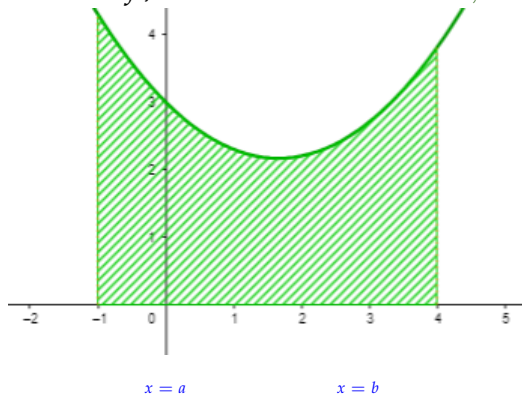
- For a rectangle, the area is the product of the length and the width.
- The area of a triangle is half the base times the height.
- The area of a polygon is found by dividing it into triangles and adding the areas of the triangles:

$$A = A_1 + A_2 + A_3 + A_4$$

However, it isn't so easy to find the area of a region with **curved sides**.

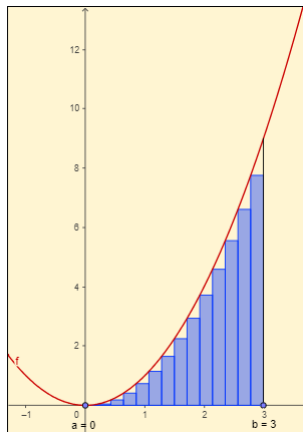
Find the area of the region that lies under the curve $y = f(x)$ on $[a, b]$

S is bounded by the graph of a continuous, **positive** function f , the **vertical lines** $x = a$, $x = b$, and the x -axis.



Area of the region that lies under the curve $y = f(x)$ on $[a, b]$: obtained by Riemann Sums

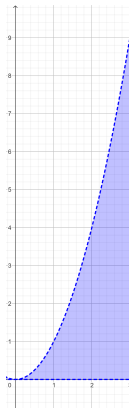
Idea: We first approximate the region S by rectangles and then we take the limit of the sum of the areas of these rectangles as we increase the number of rectangles.



Example 1.

Use N rectangles to estimate the area under the parabola $y = x^2$ from $a = 0$ to $b = 3$, for $N = 3, 6, 24$.

Exact Value: Area $= \int_0^3 x^2 dx = \frac{1}{3}x^3 \Big|_{x=0}^{x=3} = 9$



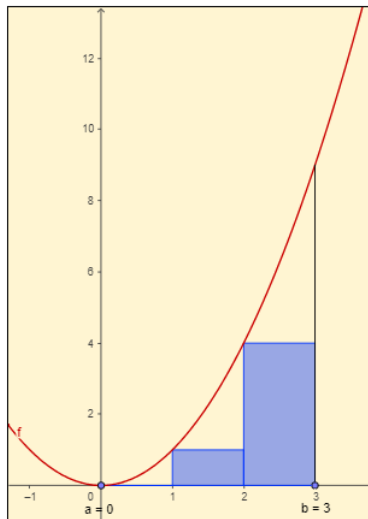
$y = x^2, a = 0, b = 3$

Example 1. Use 3 rectangles to estimate the area S under the parabola $y = x^2$ from $a = 0$ to $b = 3$.

- Divide S into 3 strips S_1, S_2, S_3 by drawing the vertical lines $x = 1, x = 2$ and $x = 3$.
 - The lower sum of the areas of the 3 rectangles of length 1 and heights $0^2, 1^2, 2^2$ is less than the area S
 - The area S is less than the upper sum of the areas of the 3 rectangles of length 1 and heights $1^2, 2^2$ and 3^2 :
 - $S_0 + S_1 + S_2 = 0 + 1 + 4 \leq \text{Area } S \leq \text{area of the upper rectangles} = 1 + 4 + 9 = 14$
 - The leftmost rectangle has collapsed because its height is 0.
 - The sum of the areas of these approximating rectangles is
- $$\text{Lower Sum} = 1 + 2^2 = 5 < \text{Area } S < 14 = \text{Upper Sum}$$

Divide S into 3 strips S_1, S_2, S_3

$$1 + 2^2 = 5 \leq \text{Area } S \leq 14$$



$n = 3$

Repeat the procedure

Use 6 rectangles to estimate the area S under the parabola $y = x^2$ from $a = 0$ to $b = 3$

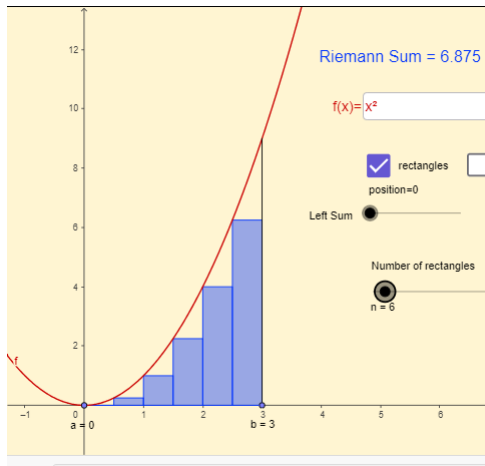
- Divide S into 6 strips $S_1, S_2, \dots, S_6$ by drawing the vertical lines $x_n = \frac{n}{2}$, $n = 0, \dots, 6$
- Each of the 6 approximating rectangles has length $\frac{1}{2}$ and heights $\left(\frac{n}{2}\right)^2$, which is value of $f(x) = x^2$ at the left endpoint of each rectangle
- The leftmost rectangle has collapsed since its height is 0.
- The area S is greater than but close to the sums of the areas of these 6 approximating rectangles, so

$$L_6 = \sum_{n=0}^5 \frac{1}{2} \cdot \left(\frac{n}{2}\right)^2 = \frac{1}{2} \left[\left(\frac{1}{2}\right)^2 + 1^2 + \left(\frac{3}{2}\right)^2 + 2^2 + \left(\frac{5}{2}\right)^2 \right]$$

$$= 6.875 \leq 9 = \text{Exact Value of the Area}$$

Divide S into 6 strips $S_1, \dots, S_6$

$\sum$ areas of 6 rectangles = $6.875 \leq \text{Area } S$



$N = 6$

Repeat the procedure

Use 24 rectangles to estimate the area S under the parabola $y = x^2$ from $a = 0$ to $b = 3$

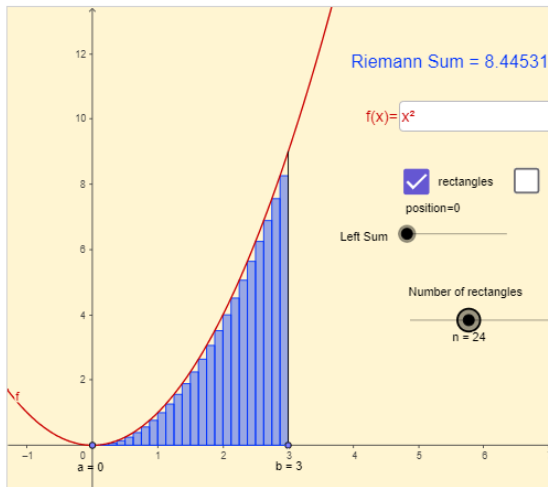
- Divide S into 24 strips $S_1, S_2, \dots, S_{24}$ by drawing the vertical lines $x_n = \frac{n}{8}, n = 0, \dots, 24$
- The area S is (greater than but) close to the sums of the areas of the 24 rectangles of length $\frac{1}{8}$ and heights $\left(\frac{n}{8}\right)^2$:

$$L_{24} = \sum_{n=0}^{23} \frac{1}{8} \left(\frac{n}{8}\right)^2 = \frac{1}{8^3} [1^2 + 2^2 + 3^2 + \dots + 23^2] = \frac{1}{8^3} \left[\frac{1}{6} 23 \cdot 24 \cdot 47 \right]$$

$$= 8.44531 \approx 9 = \text{Exact Value of Area}$$

Divide S into 24 strips $S_1, \dots, S_{24}$

$\sum$ areas of 24 rectangles = $8.44531 \approx 9 = \text{Area } S$



$N = 24$

Find the area of the region that lies under the curve $y = x^2$ on $[0, 3]$

- Instead of using smaller rectangles we could use the greater rectangles, whose heights are the values of f at the **right endpoints** of the subintervals.
- By computing the sum of the areas of the smaller rectangles L_N and the sum of the areas of the larger rectangles R_N , we obtain better lower and upper estimates for the **area**:

$$L_N < \text{Area} < R_N$$

So, the true area of S lies between $L_N < \text{Area} < R_N$

- We could obtain better estimates by **increasing N** , the number of strips.

Find the area of the region that lies under the curve $y = x^2$ on $[0, 3]$

Go the Geogebra link to see the animation on:

Approximation of the Area under the curve $y = x^2$ using rectangles, with left endpoints

- Notice that **this approximation** appears to **become better and better** as the number of strips increases, that is, as $n \rightarrow +\infty$
- Therefore, the definition of the **area A of the region S** is the **limit of the sum** of the areas of the approximating rectangles

Example 2.

Prove that the area of the region that lies under the curve $y = x^2$ on $[0, 3]$ equals $A = 9$

Alternative Proof, different of the one with Riemann Sums (to be justified later). Exact Value of the Area:

$$= \int_0^3 x^2 dx = \frac{1}{3} x^3 \Big|_{x=0}^{x=3} = 9 \checkmark$$

Ex 3. a) Prove with Riemann Sums that the area of the region under the curve $y = x^2$ on $[0, 1]$ equals $A = \frac{1}{3}$

Proof. Take limit as the amount of rectangles increases, that is,

$$\text{Area} = \lim_{n \rightarrow \infty} L_n$$

L_n is the sum of the areas of the n rectangles of width $\frac{1}{n}$ and height equal to the values of the function $f(x) = x^2$ at the left endpoints $\frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}$:

$$\Rightarrow \text{Area} = \lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \left(\frac{k}{n} \right)^2 = \lim_{n \rightarrow \infty} \left[\frac{1}{n^3} \sum_{k=1}^n k^2 \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^3} \frac{n(n+1)(2n+1)}{6}$$

$$= \lim_{n \rightarrow \infty} \frac{n^3}{6n^3} [(1 + 1/n)(2 + 1/n)] = \lim_{n \rightarrow \infty} \frac{(1 + 1/n)(2 + 1/n)}{6} = \frac{1}{3} \checkmark$$

Ex 3. b) Prove with Riemann Sums that the area of the region under the curve $y = x^2$ on $[0, 3]$ equals $A = 9$

Proof. Take limit as the amount of rectangles $n \rightarrow \infty$:

$$\text{Area} = \lim_{n \rightarrow \infty} L_n$$

L_n is the sum of the areas of the n rectangles of width $\frac{3}{n}$ and height equal to the values of $f(x) = x^2$ at the left endpoints $x_k = \frac{3k}{n}$:

Partition = $\{x_0 = 0, x_1 = \frac{1}{n}, x_2 = \frac{2}{n}, \dots, \frac{3(n-1)}{n}\}$:

$$\Rightarrow \text{Area} = \lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{3}{n} \left(\frac{3k}{n} \right)^2 = \lim_{n \rightarrow \infty} \left[\frac{27}{n^3} \sum_{k=1}^n k^2 \right]$$

$$= \lim_{n \rightarrow \infty} \frac{27}{n^3} \frac{n(n+1)(2n+1)}{6}$$

$$= \lim_{n \rightarrow \infty} \frac{27n^3}{6n^3} [(1 + 1/n)(2 + 1/n)] = \lim_{n \rightarrow \infty} \frac{9}{2} (1 + 1/n)(2 + 1/n) = 9 \quad \checkmark$$

Example 3. c)

Using antiderivatives, prove that the area of the region that lies under the curve $y = x^2$ on $[0, 3]$ equals $A = 9$

Alternative Proof, different of the one with Riemann Sums.
Exact Value of the Area:

$$= \int_0^3 x^2 dx = \frac{1}{3}x^3 \Big|_{x=0}^{x=3} = 9 \checkmark$$

Definition of Area under the curve $y = f(x)$ on $[a, b]$.

Let f be a **positive** function on $[a, b]$.

Consider a partition $\{x_k : 0 \leq k \leq n\}$ of the interval $[a, b]$ into n subintervals of equal width $\frac{b-a}{n}$.

Take the rectangles of basis equal to the length of the small intervals $[x_k, x_{k+1}]$ and height equal to the values of the function $f(x)$ at **any points** $x_k^* \in [x_k, x_{k+1}]$.

We define the area of the region under the curve $y = f(x)$ on $[a, b]$ equal to the limit of the **Riemann Sums**, that is, the limit of the sum of the areas of the approximating rectangles, as the **amount of rectangles tends to ∞** :

$$\text{Area} = \lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(b-a)}{n} f(x_k^*)$$

Definition of the Definite Integral of $f(x)$ on $[a, b]$.

Let f be a function on $[a, b]$. Divide the interval $[a, b]$ into n subintervals of equal width $\frac{b-a}{n}$, with endpoints given by

$$\{x_k : 0 \leq k \leq n\} = \{x_0 = a, x_1, x_2, \dots, x_n = b\}$$

Let $x_k^* \in [x_k, x_{k+1}]$ be any point in each subinterval

We define the definite integral of $f(x)$ on $[a, b]$ as the limit of the **Riemann Sums**, that is:

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(b-a)}{n} f(x_k^*)$$

provided that the limit exists and gives the same value for all possible choices of the sample points.

In this case, we say that f is integrable on $[a, b]$.

Theorem.

Existence of the Definite Integral of $f(x)$ on $[a, b]$.

If f is a continuous function on $[a, b]$, or f has only a finite number of jump discontinuities and each jump is of finite length, then f is integrable; that is, the definite integral

$$\int_a^b f(x)dx$$

exists.

Ex 4. Evaluate the definite integral using Riemann sums:

$$\int_0^3 (x^3 - 6x) dx$$

Solution.

- In this case, $f(x) = x^3 - 6x$ is a continuous function on $[0, 3]$, then f the definite integral exists.
- Divide the interval $[0, 3]$ into n subintervals of equal width $\frac{3}{n}$, with endpoints given by

$$\left\{ x_k = \frac{3k}{n} : 0 \leq k \leq n \right\} = \{x_0 = 0, x_1, x_2, \dots, x_n = 3\}$$

We can use any point in each subinterval $x_k^* \in [x_k, x_{k+1}]$,
so choose the left endpoints: $x_k^* = \frac{3k}{n}$

Evaluation of the definite integral using Riemann sums

$$\int_0^3 (x^3 - 6x) dx$$

- Compute the definite integral of $f(x)$ on $[0, 3]$ as the limit of the **Riemann Sums**, that is:

$$\int_0^3 (x^3 - 6x) dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{3}{n} f(x_k^*)$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \frac{3}{n} \cdot \sum_{k=0}^n \left[\left(\frac{3k}{n} \right)^3 - 6 \frac{3k}{n} \right] = \lim_{n \rightarrow \infty} \frac{3}{n} \cdot \sum_{k=0}^n \left[\frac{27}{n^3} k^3 - \frac{18}{n} k \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{81}{n^4} \cdot \sum_{k=0}^n k^3 - \frac{54}{n^2} \cdot \sum_{k=0}^n k \right] \end{aligned}$$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \left[\frac{81}{n^4} \cdot \sum_{k=0}^n k^3 - \frac{54}{n^2} \cdot \sum_{k=0}^n k \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{81}{n^4} \cdot \left[\frac{n(n+1)}{2} \right]^2 - \frac{54}{n^2} \cdot \left[\frac{n(n+1)}{2} \right] \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{81}{n^4} \cdot \frac{n^4}{4} \left(1 + \frac{1}{n} \right)^2 - \frac{54}{n^2} \cdot \frac{n^2}{2} \left(1 + \frac{1}{n} \right) \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{81}{4} \cdot \left(1 + \frac{1}{n} \right)^2 - 27 \cdot \left(1 + \frac{1}{n} \right) \right] = \frac{81}{4} - 27 = -6.75 \checkmark
\end{aligned}$$

Verify: $\int_0^3 (x^3 - 6x) dx$

$$= \left[\frac{1}{4} x^4 - 3x^2 \right] \Big|_{x=0}^{x=3} = \frac{1}{4} 3^4 - 3 \cdot 3^2 = -6.75$$

Formulas for sum of powers of integers

Consider $k, n \in \mathbb{N}$, the natural numbers, the following formulas hold:

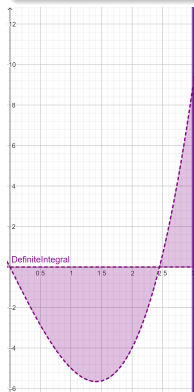
$$\sum_{k=0}^n k = \frac{n(n+1)}{2},$$

$$\sum_{k=0}^n k^2 = \frac{1}{6}n(n+1)(2n+1)$$

$$\sum_{k=0}^n k^3 = \frac{1}{4}n^2(n+1)^2$$

Evaluation of the definite integral using Riemann sums

$$\int_0^3 (x^3 - 6x)dx = -6.75$$



Notice: $\int_0^3 (x^3 - 6x)dx = -6.75$

$= -A_1 + A_2$, if A_1 is the area of the region **below** the x -axis and A_2 is the area of the region **above** the x -axis

Find an estimation of the definite integral $\int_1^2 \frac{1}{x} dx$

using Riemann sums for $n = 5$

- Consider 5 intervals so

$$\Delta x = \text{width of each subinterval} = \frac{(2-1)}{5} = \frac{1}{5},$$

- The endpoints are

$$\{x_0 = 1, x_1 = 1.2, x_2 = 1.4, x_3 = 1.6, x_4 = 1.8, x_5 = 2\}$$

- Take x_k^* equal to the middle point in each of the 5 subintervals,

$$\int_1^2 \frac{1}{x} dx \approx \frac{1}{5} \left(\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9} \right) \approx 0.6919$$

Exact Value:

$$\int_1^2 \frac{1}{x} dx = \text{Log}(2) = 0.69315$$

Properties of the definite integral

Let f, g be integrable functions on an interval $[a, b]$, then

$$\textcircled{1} \quad \int_a^a f(x) \, dx = 0$$

$$\textcircled{2} \quad \int_b^a f(x) \, dx = - \int_a^b f(x) \, dx$$

$$\textcircled{3} \quad \int_a^b c \, dx = c(b - a)$$

$$\textcircled{4} \quad \int_a^b [f(x) + g(x)] \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx$$

$$\textcircled{5} \quad \int_a^b c \cdot f(x) \, dx = c \cdot \int_a^b f(x) \, dx, \quad c \in \mathbb{R}$$

$$\textcircled{6} \quad \int_a^b [f(x) - g(x)] \, dx = \int_a^b f(x) \, dx - \int_a^b g(x) \, dx$$

Geometrical Property

Suppose that $a < c < b$ then

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$$

Comparison Properties of the definite integral

Let f, g be integrable functions on an interval $[a, b]$, then

- ① If $f \geq 0$ on $[a, b]$, then $\int_a^b f(x) dx \geq 0$ and the integral represents the area below the curve $y = f(x)$.

- ② If $f \geq g$ on $[a, b]$ then

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx$$

- ③ If $m \leq f \leq M$ on $[a, b]$ then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$